



SCHOOL OF COMPUTING AND ENGINEERING SCIENCES
BACHELOR OF SCIENCE IN INFORMATICS AND COMPUTER SCIENCE (BICS)
CAT 1
ICS 1103: Differential Calculus

DATE: 9th August 2024

TIME: 1 Hour

Instructions: Answer ALL questions. Show your workings.

✓ (a) Given $f(x) = -5 - 4x$ and $g(x) = x - 7$, find

(i) $(g \circ f)(x)$ [3 Marks]

(ii) $(g \circ f)(10)$ [2 Marks]

(b) Determine the inverse of the function $h(x)$, where [3 Marks]

$$h(x) = 7(2x - 5)$$

(c) A peach grower takes steps to improve the quality of his product over several years. The annual profits follow the function

$$P(t) = \frac{25(t + 4)}{t + 5}$$

where t is the time in years and $P(t)$ is profit in thousands of dollars.

(i) Draw the graph of P in the viewing window $[0,10]$ by $[0,30]$ [3 Marks]

(ii) Find the annual profit 3 years after the start of the improvement program [3 Marks]

(iii) Find and interpret $\lim_{t \rightarrow \infty} P(t)$. [2 Marks]

✓ (d) The resale value R (in dollars) of a certain type of industrial equipment has been found to behave according to the function $R(t) = 900,000e^{-0.02t}$, where t is the number of years since original purchase.

(i) Determine the original value of the piece of equipment. [2 Marks]

(ii) Determine the number of years it will take for the value of the piece of equipment to reach \$ 600,000. [3 Marks]



SCHOOL OF COMPUTING AND ENGINEERING SCIENCES
BACHELOR OF BUSINESS INFORMATION TECHNOLOGY (BBIT)

Continuous Assessment Test 1

BBT 2104 : Differential Calculus (Calculus 1)

Date: 26th September, 2023

Time: 1 Hour

Attempt ALL questions and show your workings

1. Determine the indicated limits.

(i) $\lim_{h \rightarrow 0} (x + 2h)^2$ [2 Marks]

(ii) $\lim_{x \rightarrow -1} \frac{x + 1}{|x + 1|}$ [2 Marks]

(iii) $\lim_{x \rightarrow 3} \left(\frac{x^2 - 3x}{x^2 - 9} \right)$ [2 Marks]

(iv) $\lim_{x \rightarrow 2} \left(\frac{\sqrt{x} - 2}{x - 4} \right)$ [2 Marks]

2. State the three conditions that a function $f(x)$ should satisfy to be continuous at a point $x = a$, where a is any real number. [3 Marks]

3. Determine the continuity of the given function at the indicated point [2 Marks]

$$f(x) = \begin{cases} x - 3, & x \leq 0 \\ x^2 + x - 3, & x > 0, \end{cases} \quad \text{at } x = 0$$

4. The number of dollars spent on advertising for a product influences the number of items of the product that will be purchased by consumers. If x is the number of dollars spent, in thousands, then $N(x)$, in hundreds, is the number of items sold, and is given by the function

$$N(x) = 3200 - \frac{750}{x}$$

(a) Determine $\lim_{x \rightarrow \infty} N(x)$ and interpret. [2 Marks]

(b) Draw the graph of N in an appropriate viewing window. From the graph, estimate an interval for x that would result in $N(x)$ being between 2700 and 3000. [4 Marks]

A function f is defined by

$$f(x) = x^2 + 1, \quad x \in \mathbb{R}, \quad x \geq 0$$

(a) Find the expression for $f^{-1}(x)$.

[2 Marks]

(b) State the domain and range of $f^{-1}(x)$.

[2 Marks]

Figure 1 shows the graph of

$$y = \frac{1}{x} + 2, \quad x \neq 0$$

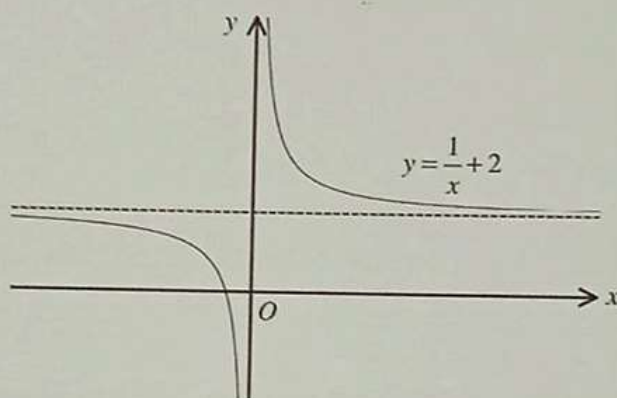


Figure 1: Graph of $y = f(x)$.

State the equation of the horizontal asymptote to the curve marked as a dotted line in the figure 1 [1 Mark]

A function f is defined by

$$f(x) = \frac{1}{2}e^x + 1, \quad x \in \mathbb{R}, \quad x \leq 0$$

(a) Find the expression for $f^{-1}(x)$.

[2 Marks]

(b) State the domain and range of $f^{-1}(x)$.

[2 Marks]

8. Let x be the proportion of the total popular vote that a Democratic candidate for president receives in a U.S national election. Political scientists have observed that a good estimate of the proportion of seats in the House of representatives going to Democratic candidates is given by the function

$$f(x) = \frac{x^3}{x^3 + (1-x)^3} \quad 0 \leq x \leq 1,$$

whose domain is the interval $[0, 1]$. Compute $f(0.62)$ and interpret the result.

[3 Marks]

9. The resale value R (in dollars) of a certain type of industrial equipment has been found to behave according to the function $R(t) = 25000e^{-0.03t}$, where t is the number of years since original purchase.

(i) Determine the original value of the piece of equipment.

[2 Marks]

(ii) Determine the value of a piece of equipment after 8 years.

[2 Marks]

END
